Scalability & Future Plan

Date	15 March 2026
Team ID	SWTID-2026-4536
Project Name	LegalEase – AI-Powered Legal Document Generator
Maximum Marks	1 Mark

Scalability & Future Plan

This document outlines how the current LegalEase solution can be scaled to handle larger user bases, increased document generation requests, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the LegalEase project beyond its current state.

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	Dependency on Gemini AI API for legal document generation	Document generation depends on API availability and configured API access.	**High**
2	Local deployment using FastAPI and Streamlit	Large numbers of simultaneous users may require additional deployment and infrastructure support.	**Medium**
3	Limited document types and legal-domain coverage	The current system focuses on common legal documents such as contracts, NDAs, lease agreements, and	**Medium**

		employment documents.	

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	**User Load**	LegalEase currently uses Streamlit as the frontend and FastAPI as the backend for document generation.	Deploy the application on scalable cloud infrastructure and support a larger number of concurrent users.
2	**Data Storage**	Generated documents and user inputs are handled during the document generation process.	Introduce secure and scalable database/storage solutions for managing documents and user data.
3	**Performance**	Gemini AI generates legal content through the FastAPI backend and the generated output is displayed through Streamlit.	Optimize API processing, document generation, and deployment infrastructure to improve response time and support increased workloads.

4	**Security**	The project uses environment variables such as a `.env` file for securely storing the Gemini API key.	Strengthen data privacy and security mechanisms for sensitive legal information and secure API access.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Improve the existing LegalEase document generation, editing, preview, and export functionality.	**Short Term**	Improved reliability and user experience.
Phase 2	Add deeper contract analysis, legal text analysis, and enhanced document understanding capabilities.	**Medium Term**	Better understanding and analysis of legal documents.
Phase 3	Integrate legal databases and expand the range of supported legal documents and use cases.	**Long Term**	Broader legal coverage and improved usefulness for users.

Phase 4	Introduce personalized recommendations and enhanced AI-powered legal assistance with scalable deployment.	**Long Term**	More personalized, accessible, and scalable legal document assistance